Proposal of Modelling and Architectural Improvements for Brain-to-Text Decoding

Motivation

Our research originates from a Kaggle competition (Brain-to-text '25 | Kaggle) dedicated to translating brain signals into natural language, which itself is based on foundational work by Card et al. (2024) published in *The New England Journal of Medicine*. After a thorough analysis of the paper's methodologies and the provided code. We have identified three primary directions for improvement: reconstructing the acoustic model architecture, enhancing language-model fusion, and explore the possibility of implementing an end-to-end architecture. We plan to deliver robust benchmarks and modular software artefacts to the speech-BCI research community.

Related Work

The reference paper (Card et al., 2024) employs a two-stage architecture for brain-to-text decoding, where an acoustic model first maps cortical neural activity to phoneme probabilities, followed by a language model that converts phonemes into text. The acoustic decoder is based on recurrent neural networks trained with a Connectionist Temporal Classification (CTC) objective, while the language stage uses n-gram or Transformer-based models, with subsequent studies incorporating LLMs for sentence-level rescoring. This two-stage framework provides the methodological foundation for our proposed extensions.

In language modelling, Bradbury et al. (2017) proposed the Quasi-Recurrent Neural Network (QRNN), combining convolutional and recurrent operations to achieve high parallelism and faster training compared to LSTMs. Katharopoulos et al. (2020) introduced linear attention, reducing the quadratic complexity of standard Transformers to linear time and memory while maintaining strong modeling performance. Hinton et al. (2015) developed Knowledge Distillation, enabling smaller student models to learn from larger teacher models and achieve high efficiency with minimal accuracy loss.

Beyond language modelling, recent advances in brain-to-text decoding have also explored end-to-end architectures which can map brain signals directly to text. They have been used to decode brain signals with RNNs or CNNs (Sun et al., 2020; Makin et al., 2020), and more recently with Transformer encoder–decoder models (Komeiji et al., 2024). Research on Transformer-based brain-to-text decoding remains limited.

Additionally, ensemble methods are increasingly central to BCI-to-Text decoding. Most systems generate multiple candidate sentences and apply LLM-based rescoring to achieve substantial error-rate reductions (Willett et al., 2024), whereas classical fusion strategies such as confidence-weighted or majority voting (ROVER) continue to yield competitive results without relying on LLMs (Parikh et al., 2024).

Description of Proposal

The Transducer architecture extends CTC by defining a distribution over output sequences of all lengths, and by jointly modelling both input-output and output-output dependencies (Graves, 2012). Therefore, we propose to use the Transformer Transducer architecture for the acoustic model which was proposed by Google (Zhang et al., 2020) for ASR tasks. This model replaces the traditional RNN encoders with Transformer blocks and uses the Transducer loss, defined as the

negative log- likelihood of the correct output sequence given the input sequence. Compared with the RNN-T, the Transformer Transducer architecture offers two major advantages: it enables flexible latency control by adjusting the self-attention mask to include a limited look-ahead window, and it allows faster training through the parallel computation inherent in the self-attention mechanism.

Additionally, we will also consider the benefit of leveraging the Conformer module introduced by Google (Gulati et al., 2020). Recent research has begun to apply this architecture for decoding electroencephalograph (EEG) signals. (Song et al., 2023; Wang et al., 2025)

Current Transformer-based language models, though highly effective, face several limitations in brain-to-text decoding. Therefore, we propose three possible ways to improve the efficiency of the language model. First, we consider QRNNs, which offer greater parallelism and faster computation (Bradbury et al., 2017). Second, we plan to adopt linear-attention mechanisms such as Performer or Linformer, which can reduce computational complexity from $O(L^2)$ to $O(L)$ (Katharopoulos et al., 2020). Third, since large Transformer models remain computationally heavy, we will apply knowledge distillation to compress them into lightweight student networks that preserve performance while accelerating inference (Hinton et al., 2015).

In terms of end-to-end architectures, inspired by the success of the Transformer Transducer model in speech-to-text applications (Yeh et al., 2019), we propose to investigate whether this architecture can achieve efficient and accurate brain-to-text decoding.

Finally, we propose two complementary sentence-level ensembling strategies for BCI-to-Text decoding. First, a staged approach uses one acoustic model to generate phonemes, decoded by multiple language models with large language models scoring and selecting the best candidate. Second, we explore LLM-free fusion via exact-match voting and confidence-weighted selection—choosing either the most frequent sentence or weighting candidates by model performance or confidence. Both methods are modular and can be integrated without retraining the acoustic model.

Details of Implementation and Timeline

Phase 1 (about 6 weeks) - Acoustic Model Baseline and Transformer Transducer

Build an acoustic model baseline by fine-tuning a pre-trained 2024 RNN model on BCI data and exploring a Transformer Transducer model to replace traditional CTC. Deliverables include the RNN and T-T baselines along with their respective performance reports.

Phase 2 (about 4 weeks) - Efficient Language Model Development

Train and compare three efficient LM variants: a QRNN model for faster computation, a Performer model with linear attention for long sequences, and a DistilBERT model via knowledge distillation. Deliverables include the trained models and a corresponding efficiency analysis.

Phase 3 (about 4 weeks) - Sentence-Level Fusion Decoding

Implement two decoding fusion strategies: (1) LLM-based re-ranking using LLaMA/GPT, and (2) lightweight fusion via voting or confidence weighting. Deliverables include the decoding modules and a detailed Word Error Rate (WER) analysis.

Phase 4 (about 2 weeks) - Evaluation and Final Report

Evaluate all components on performance, efficiency, and cost. Summarize findings and improvements in the final project report. Deliverables include submitting the comprehensive evaluation and the final project report.

Reference

Bradbury, J., Merity, S., Xiong, C., & Socher, R. (2016). Quasi-recurrent neural networks. *arXiv preprint arXiv:1611.01576*.

Card, N. S., Wairagkar, M., Iacobacci, C., Hou, X., Singer-Clark, T., Willett, F. R., ... & Brandman, D. M. (2024). An accurate and rapidly calibrating speech neuroprosthesis. *New England Journal of Medicine*, *391*(7), 609-618.

Graves, A. (2012). Sequence transduction with recurrent neural networks. *arXiv preprint arXiv:1211.3711*.

Hinton, G., Vinyals, O., & Dean, J. (2015). Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*.

Katharopoulos, A., Vyas, A., Pappas, N., & Fleuret, F. (2020, November). Transformers are rnns: Fast autoregressive transformers with linear attention. In *International conference on machine learning* (pp. 5156-5165). PMLR.

Komeiji, S., Mitsuhashi, T., Iimura, Y., Suzuki, H., Sugano, H., Shinoda, K., & Tanaka, T. (2024). Feasibility of decoding covert speech in ECoG with a Transformer trained on overt speech. *Scientific Reports*, *14*(1), 11491.

Makin, J. G., Moses, D. A., & Chang, E. F. (2020). Machine translation of cortical activity to text with an encoder–decoder framework. *Nature neuroscience*, *23*(4), 575-582.

Parikh, A. K., ten Bosch, L., & van den Heuvel, H. (2024, May). Ensembles of Hybrid and End-to-End Speech Recognition. In *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)* (pp. 6199-6205).

Sun, P., Anumanchipalli, G. K., & Chang, E. F. (2020). Brain2Char: a deep architecture for decoding text from brain recordings. *Journal of neural engineering*, *17*(6), 066015.

Willett, F. R., Li, J., Le, T., Fan, C., Chen, M., Shlizerman, E., ... & Henderson, J. M. (2024). Brain-to-Text Benchmark'24: Lessons Learned. *arXiv preprint arXiv:2412.17227*.

Yeh, C. F., Mahadeokar, J., Kalgaonkar, K., Wang, Y., Le, D., Jain, M., ... & Seltzer, M. L. (2019). Transformer-transducer: End-to-end speech recognition with self-attention. *arXiv preprint arXiv:1910.12977*.

Zhang, Q., Lu, H., Sak, H., Tripathi, A., McDermott, E., Koo, S., & Kumar, S. (2020, May). Transformer transducer: A streamable speech recognition model with transformer encoders and rnn-t loss. In *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)* (pp. 7829-7833). IEEE.

Gulati, A., Qin, J., Chiu, C.-C., Parmar, N., Zhang, Y., Yu, J., ... & Pang, R. (2020). Conformer: Convolution-augmented transformer for speech recognition.

Song, Y., Zheng, Q., Liu, B., & Gao, X. (2023). EEG Conformer: Convolutional Transformer for EEG Decoding and Visualization. *In 2023 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*

Wang, Z., Wang, H., Jia, T., He, X., Li, S., & Wu, D. (2025). DBConformer: Dual-Branch Convolutional Transformer for EEG Decoding. IEEE Journal of Biomedical and Health Informatics.